Visualization Basics for Survey Data

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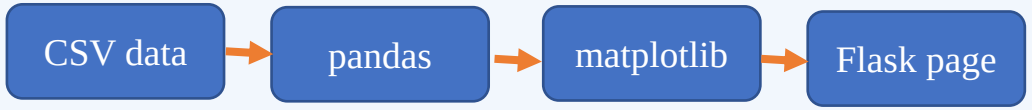
WEEK 4 LECTURE DECK

Visualization Basics for Survey Data

Topics this week:

- choosing the right chart
- matplotlib basics: bar, line, histogram
- chart formatting and saving figures
- using visuals in a simple Flask dashboard

Sample project
Online Survey System



Survey data becomes a picture people can understand.

This week, students learn how to turn cleaned data into clear charts. The goal is not only to make a chart, but to choose the chart that matches the question.

Week 4 roadmap

What students should understand and do by the end of this week

1 Choose chart

Match the chart to the question

2 Bar chart

Compare categories

3 Line chart

See change over time

4 Histogram

See score distribution

5 Format charts

Title, labels, legend

6 Save figure

Use image in Flask page

Tip: first choose the right chart, then make the chart look clean and readable.

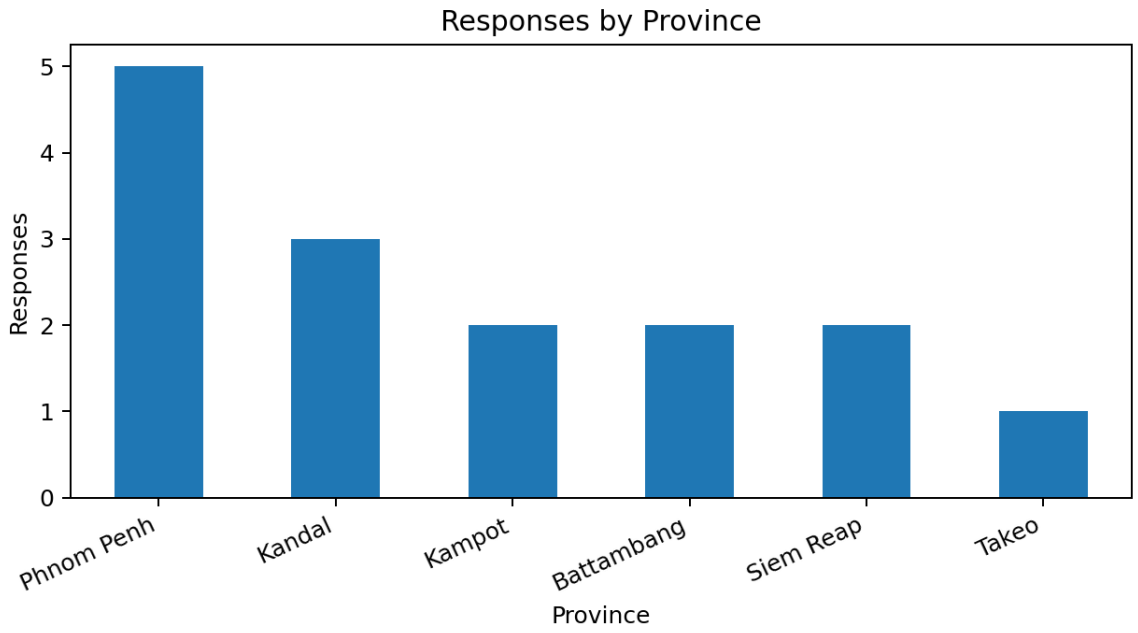
Why do we use charts?

A chart should make the message faster and clearer

Table only

Province	Avg score
Phnom Penh	4.0
Kandal	4.5
Siem Reap	1.5
Takeo	2.0

The numbers are correct, but the weak province is not easy to notice at first look.



With a chart, we can quickly compare categories and spot strong or weak areas.

Choose the right chart for the question

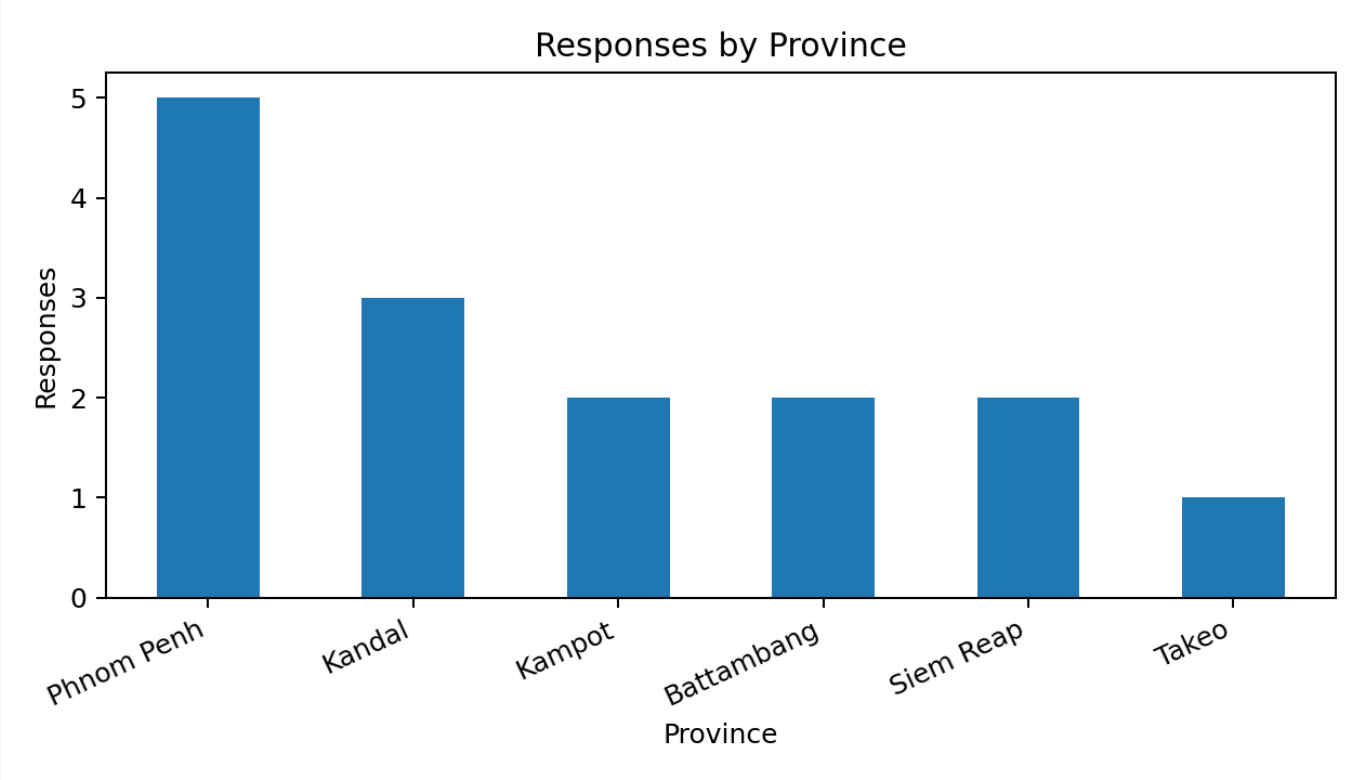
Simple rule: the question tells you the chart type

Question type	Good chart	Survey example
Compare categories	**Bar chart**	Responses by province or device
See change over time	**Line chart**	Average score by day or week
See value distribution	**Histogram**	How many users gave score 1, 2, 3, 4, 5
Show part of whole	**Pie chart (use carefully)**	Share of device type when there are few groups

Bad practice to avoid: choosing a chart because it looks nice, not because it answers the question.

Bar chart: best for category comparison

Session 1 example from the online survey system



- Use a bar chart when the x-axis has categories, not dates.
- Each bar shows one group, such as province, device, or age group.
- Bar height shows the value: count, average, or total.
- Keep category labels short and easy to read.

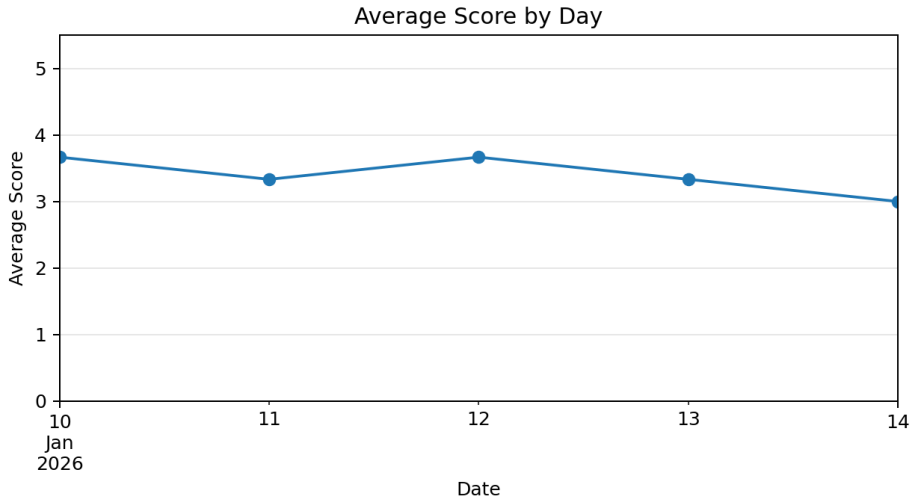
Reading tip

Ask: Which bar is tallest? Which bar is shortest? That helps students read the chart with meaning, not only with code.

Line chart and histogram

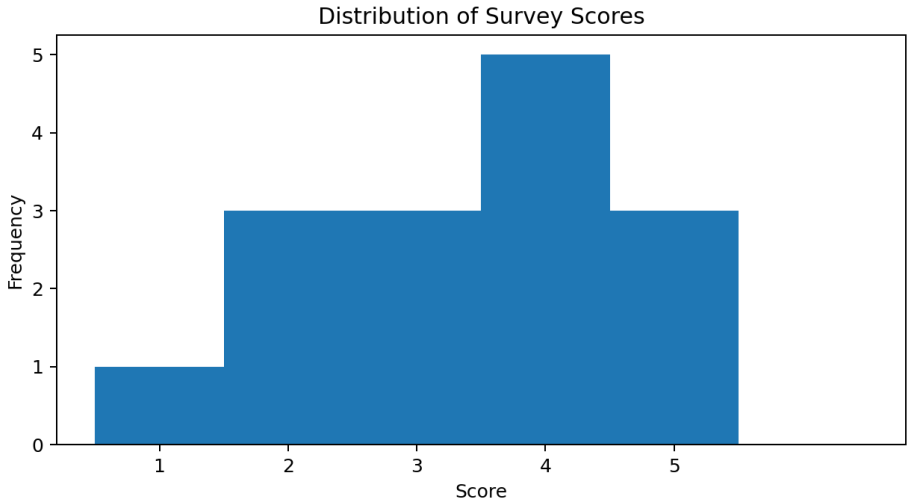
Two different questions, two different chart jobs

Line chart = change over time



Use a line chart when the x-axis is day, week, month, or another time order.

Histogram = distribution of values



Use a histogram when you want to see how values are spread, such as many low scores or many high scores.

Session 2: chart formatting rules

Make the chart easier to read, not more complicated

- Use a clear title that says what the chart shows.
- Label both axes when the meaning is not obvious.
- Use a legend only when it helps the reader.
- Rotate long labels only if needed.
- Keep extra colors, lines, and effects to a minimum.

Good chart checklist

✓ **Title**	Clear and specific
✓ **Axis labels**	Easy to understand
✓ **Legend**	Only when needed
✓ **Font size**	Readable from the back of the room
✓ **Output file**	Saved with a clear name

Matplotlib code example

Bar chart from survey data in simple Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("survey_responses.csv")
summary = df.groupby("province")["response_id"].count()

summary.plot(kind="bar")
plt.title("Responses by Province")
plt.xlabel("Province")
plt.ylabel("Responses")
plt.tight_layout()
plt.savefig("responses_by_province.png")
plt.show()
```

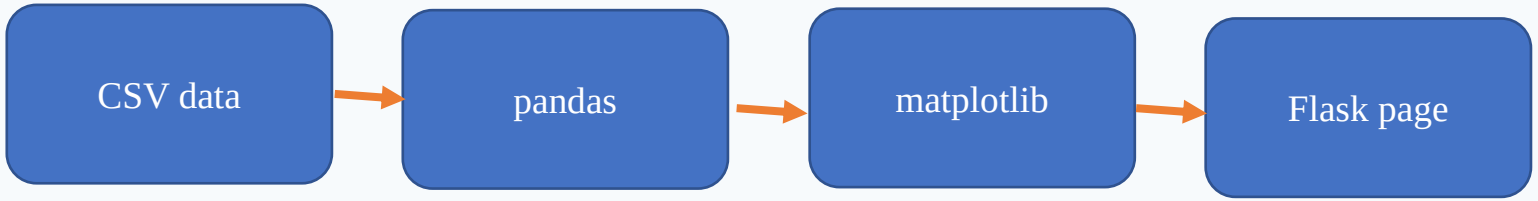
- read_csv loads the file.
- groupby + count makes the category summary.
- plot(kind="bar") creates the chart.
- savefig saves the chart for reports or Flask.

Tip:

Run the code line by line and ask students to predict the output before they see the image.

From matplotlib to Flask page

A saved figure can become part of the dashboard



Save the image

```
plt.savefig("static/charts/responses_by_province.png")
```

Now Flask can show the file inside an HTML page.

HTML example

```
<img src="/static/charts/responses_by_province.png"
    alt="Responses by Province">
```

This is the simple path from analysis to dashboard view.

Common visualization mistakes

Students should learn what to avoid

Mistake

- ✗ Too many colors or effects
- ✗ Wrong chart for the question
- ✗ Long titles and unclear labels
- ✗ Unreadable font size
- ✗ No title, no units, no context

Better practice

- ✓ Use simple, limited color choices
- ✓ Start from the business question
- ✓ Write short labels in plain words
- ✓ Check readability from distance
- ✓ Always tell what, where, and when

Week 4 class activity and recap

Finish the week with one small charting success

In-class activity

- Load the survey CSV file.
- Make one bar chart by province or device.
- Add title, x-label, and y-label.
- Save the figure as a PNG file.
- Explain one insight in one sentence.

By the end of Week 4

- Students know when to use bar, line, and histogram.
- Students can make basic matplotlib charts.
- Students can format and save a chart image.
- Students are ready to place charts into Flask dashboards.

Remember: the best chart is the one people understand quickly.

app2.py

```python
import os
from flask import Flask, render_template
import pandas as pd
import matplotlib

# Use non-GUI backend for server/chart saving
matplotlib.use("Agg")
import matplotlib.pyplot as plt

app = Flask(__name__)

CHART_FOLDER = os.path.join("static", "charts")

def ensure_chart_folder():
    os.makedirs(CHART_FOLDER, exist_ok=True)

def load_data():
    df = pd.read_csv("survey_responses.csv")

    # Basic cleaning
    df["province"] = df["province"].astype(str).str.strip().str.title()
    df["age_group"] = df["age_group"].astype(str).str.strip()
    df["device"] = df["device"].astype(str).str.strip().str.lower()
    df["recommend"] = df["recommend"].astype(str).str.strip().str.lower()
    df["score"] = pd.to_numeric(df["score"], errors="coerce")
    df["submit_time"] = pd.to_datetime(df["submit_time"], errors="coerce")

    return df

def create_bar_chart(df):
    province_counts = df["province"].value_counts().sort_index()

    plt.figure(figsize=(10, 5))
    province_counts.plot(kind="bar")
    plt.title("Responses by Province")
    plt.xlabel("Province")
    plt.ylabel("Number of Responses")
    plt.xticks(rotation=30, ha="right")
    plt.tight_layout()
    plt.savefig(os.path.join(CHART_FOLDER, "responses_by_province.png"))
    plt.close()

def create_line_chart(df):
    device_avg = (
        df.groupby("device")["score"]
        .mean()
        .round(2)
        .sort_index()
    )

    plt.figure(figsize=(8, 5))
    plt.plot(device_avg.index, device_avg.values, marker="o")
    plt.title("Average Score by Device")
    plt.xlabel("Device")
    plt.ylabel("Average Score")
    plt.ylim(0, 5.5)
    plt.grid(True)
    plt.tight_layout()
    plt.savefig(os.path.join(CHART_FOLDER, "avg_score_by_device.png"))
    plt.close()

def create_histogram(df):
    plt.figure(figsize=(8, 5))
    plt.hist(df["score"].dropna(), bins=5, edgecolor="black")
    plt.title("Score Distribution")
    plt.xlabel("Score")
    plt.ylabel("Frequency")
    plt.tight_layout()
    plt.savefig(os.path.join(CHART_FOLDER, "score_distribution.png"))
    plt.close()

def create_summary_tables(df):
    province_summary = (
        df.groupby("province")
        .agg(
            total_responses=("response_id", "count"),
            average_score=("score", "mean")
        )
        .reset_index()
    )
    province_summary["average_score"] = province_summary["average_score"].round(2)
    province_summary = province_summary.sort_values(
        by="average_score", ascending=False
    )

```

```python
    age_summary = (
        df.groupby("age_group")
        .agg(
            total_responses=("response_id", "count"),
            average_score=("score", "mean")
        )
        .reset_index()
    )
    age_summary["average_score"] = age_summary["average_score"].round(2)

    return province_summary, age_summary

@app.route("/")
def index():
    ensure_chart_folder()
    df = load_data()

    # KPI values
    total_responses = len(df)
    average_score = round(df["score"].mean(), 2)
    min_score = df["score"].min()
    max_score = df["score"].max()

    # Create charts
    create_bar_chart(df)
    create_line_chart(df)
    create_histogram(df)

    # Tables
    province_summary, age_summary = create_summary_tables(df)

    province_table = province_summary.to_html(
        classes="table table-bordered table-striped",
        index=False
    )

    age_table = age_summary.to_html(
        classes="table table-bordered table-striped",
        index=False
    )

    # Simple insight
    top_province = province_summary.iloc[0]["province"]
    top_score = province_summary.iloc[0]["average_score"]
    insight = (
        f"The highest average score is in {top_province} "
        f"with a score of {top_score}."
    )

    return render_template(
        "index_v2.html",
        total_responses=total_responses,
        average_score=average_score,
        min_score=min_score,
        max_score=max_score,
        province_table=province_table,
        age_table=age_table,
        insight=insight
    )

if __name__ == "__main__":
    app.run(debug=True)
```

templates > index_v2.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Week 4 Survey Dashboard</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 30px;
            background-color: #f4f6f9;
        }

        h1, h2 {
            color: #1f3c88;
        }

        .card {
            background: white;
            border-radius: 10px;
            padding: 20px;
            margin-bottom: 20px;
            box-shadow: 0 2px 8px rgba(0,0,0,0.08);
        }

        .kpi-container {
            display: flex;
            gap: 20px;
            flex-wrap: wrap;
        }

        .kpi-box {
            flex: 1;
            min-width: 180px;
            background: white;
            border-radius: 10px;
            padding: 20px;
            text-align: center;
            box-shadow: 0 2px 8px rgba(0,0,0,0.08);
        }

        .kpi-box h3 {
            margin: 0;
            font-size: 16px;
            color: #444;
        }

        .kpi-box p {
            font-size: 28px;
            margin-top: 10px;
            font-weight: bold;
            color: #1f3c88;
        }

        img {
            max-width: 100%;
            border-radius: 8px;
            border: 1px solid #ddd;
            margin-top: 10px;
        }

        table {
            width: 100%;
            border-collapse: collapse;
            background: white;
        }

        th, td {
            border: 1px solid #ccc;
            padding: 10px;
            text-align: center;
        }

        th {
            background-color: #dbeafe;
        }
    </style>
</head>
```

templates/index_v2.html

```html

<body>
    <h1>Online Survey System - Week 4 Dashboard</h1>

    <div class="card">
        <h2>Summary KPI</h2>
        <div class="kpi-container">
            <div class="kpi-box">
                <h3>Total Responses</h3>
                <p>{{ total_responses }}</p>
            </div>
            <div class="kpi-box">
                <h3>Average Score</h3>
                <p>{{ average_score }}</p>
            </div>
            <div class="kpi-box">
                <h3>Minimum Score</h3>
                <p>{{ min_score }}</p>
            </div>
            <div class="kpi-box">
                <h3>Maximum Score</h3>
                <p>{{ max_score }}</p>
            </div>
        </div>
    </div>

    <div class="card">
        <h2>Bar Chart: Responses by Province</h2>
        <img src="{{ url_for('static', filename='charts/responses_by_province.png') }}" alt="Responses by Province">
    </div>

    <div class="card">
        <h2>Line Chart: Average Score by Device</h2>
        <img src="{{ url_for('static', filename='charts/avg_score_by_device.png') }}" alt="Average Score by Device">
    </div>

    <div class="card">
        <h2>Histogram: Score Distribution</h2>
        <img src="{{ url_for('static', filename='charts/score_distribution.png') }}" alt="Score Distribution">
    </div>

    <div class="card">
        <h2>Province KPI Table</h2>
        {{ province_table|safe }}
    </div>

    <div class="card">
        <h2>Age Group KPI Table</h2>
        {{ age_table|safe }}
    </div>

    <div class="card">
        <h2>Quick Insight</h2>
        <p>{{ insight }}</p>
    </div>

</body>
</html>
```

Result:

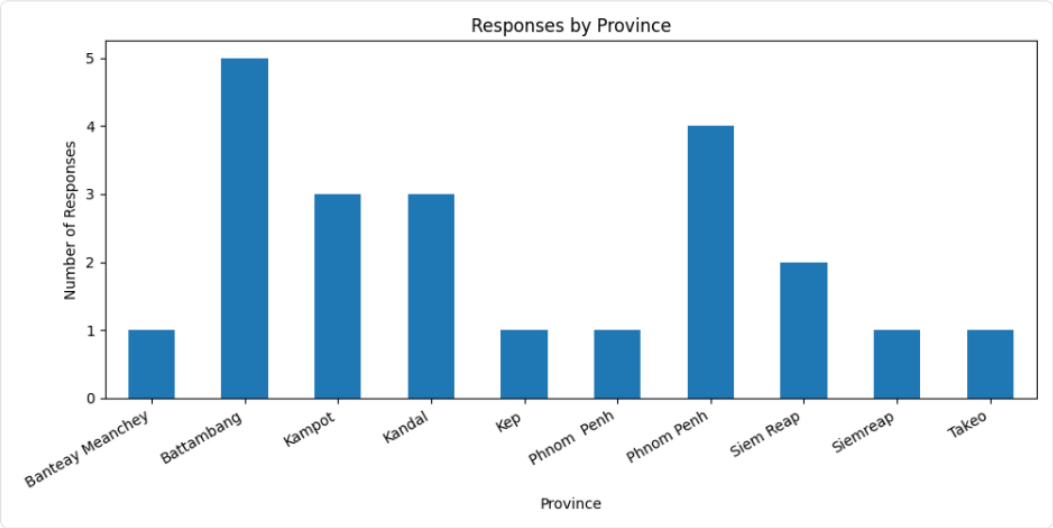
127.0.0.1:5000

Online Survey System - Week 4 Dashboard

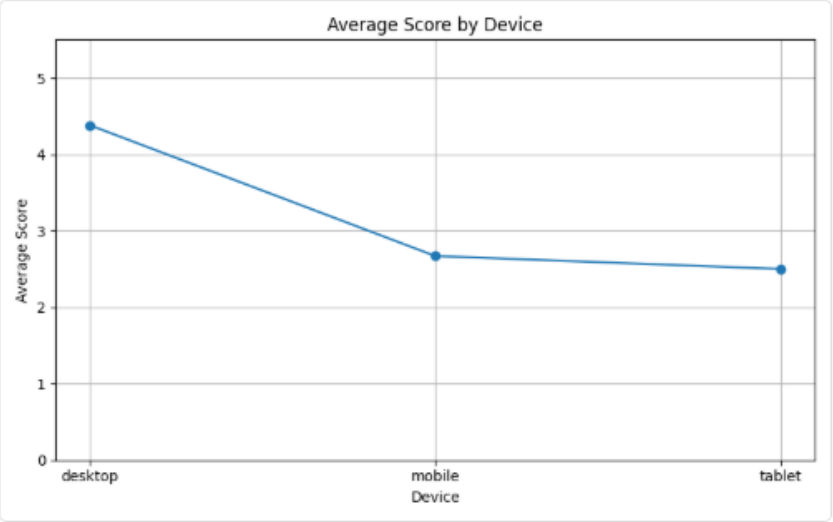
Summary KPI

Total Responses	Average Score	Minimum Score	Maximum Score
22	3.21	-1.0	6.0

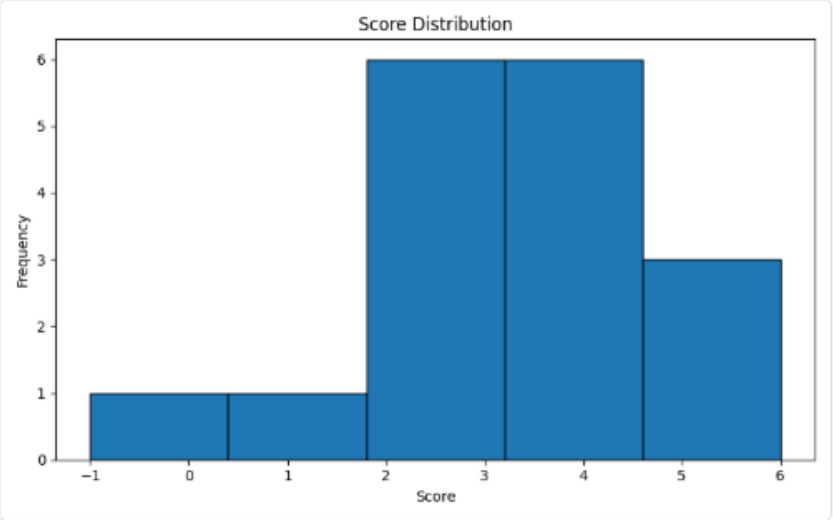
Bar Chart: Responses by Province



Line Chart: Average Score by Device



Histogram: Score Distribution



Province KPI Table

province	total_responses	average_score
Takeo	1	6.00
Phnom Penh	4	4.00
Kep	1	4.00
Kampot	3	3.83
Kandal	3	3.67
Battambang	5	2.00
Siem Reap	2	2.00
Siemreap	1	1.00
Phnom Penh	1	-1.00
Banteay Meanchey	1	NaN

Age Group KPI Table

age_group	total_responses	average_score
18-24	8	2.75
24-18	1	NaN
25-34	5	4.00
25_34	1	3.00
35-44	5	3.50
45-54	1	2.00

Quick Insight

The highest average score is in Takeo with a score of 6.0.